Multimedia

Presentation of lectures

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Lecture topic

Image and video compression

Motto

*„Is it possible to reduce the image data size?
How is it done and what are the features?“*

Contents:

- Compression classifications
- Simple image compressions
- Image specific compression
- Image compression implementation
- Video compression
- Video compression implementation

Compression classification

Classification of image information to compress

- General computer data
- Static images (specific image redundancy)
  - what is this???
- Video sequences (spatial and temporal redundancy)

Classification of image compression types

- Loseless compression (binary redundancy only)
- Lossy compression (also specific redundancy)

Simple image compressions

Loseless static image compressions

- Standard methods used also elsewhere:
  Hufmann, arithmetic, dictionary based encoding, etc,.
  (may behave well on synthesized images but do not work for photogaph type images – explained further)
- Image specific used seldom elsewhere:
  Run Length Encoding (RLE)

  FAX Encoding (CCITT)

RLE: 555556 → 6x5

FAX: Hufmann probablity based

Image specific compression

Lossy static image compression

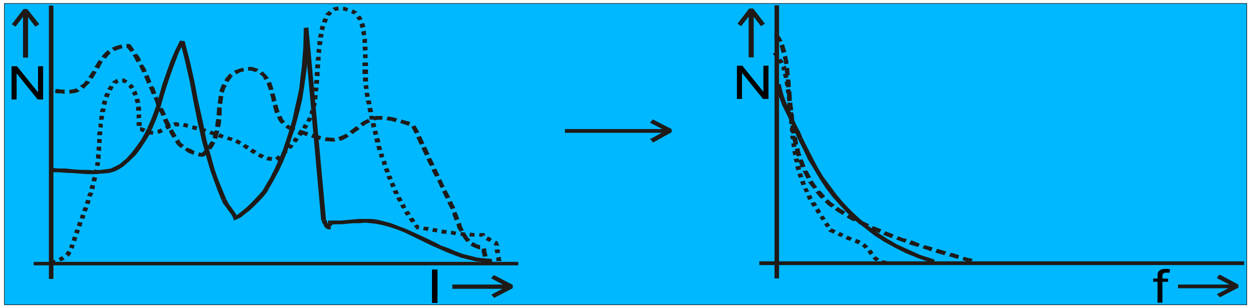
- Frequency transform – DCT or Fourier transformation based methods (more or less local)
- Wavelet type transforms (various bases) – local frequency transformation in its nature
- Other methods of search for visual redundancy, such as fractal based methods

Lossy compression might in some case be affected with visually noticeable artefacts

Image specific compression

Sources of compression:

- Most of the methods are based on the fact that distribution of the values in intensity (colour) domain is unpredictable but in frequency domain quite predictable



- In lossy compression, another source is human insensitivity to some spatial frequencies

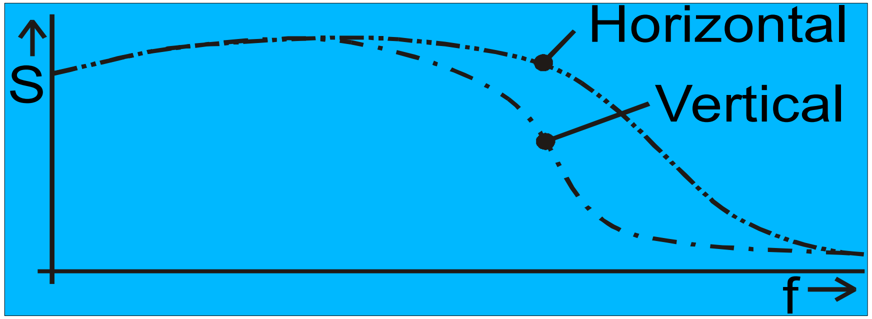


Image specific compression

Sources of compression in video (lossy):

- In addition to image sources of compression, video might also use temporal features of data and human vision

- Temporal coherence between the adjacent images in video exists – content of video frames can be quite well predicted using the “close in time” images
- The video compression can often be helped by identification of objects shown in video and their high-level animation
- Behaviour of humans when watching video sequence is (at least to large extent) predictable – this fact can be exploited by defining the more and less important regions

Image compression implementation

Typical example of image compression algorithm based on frequency transform is defined by JPEG (Joint Picture Encoding Group); it is based on the local frequency transform (DCT on 8x8 pixels blocks) and probability based (Hufmann) encoding, the more detailed steps are the following (first, grayscale version is explained):

- DCT is applied on each of the image's 8x8 pixel blocks
- Sensitivity correction (based on sight features) is applied
- Huffmann encoding is performed under control of the quality or size measures

- The decompression process is merely an inversion of the above steps

Image compression implementation

JPEG Scheme

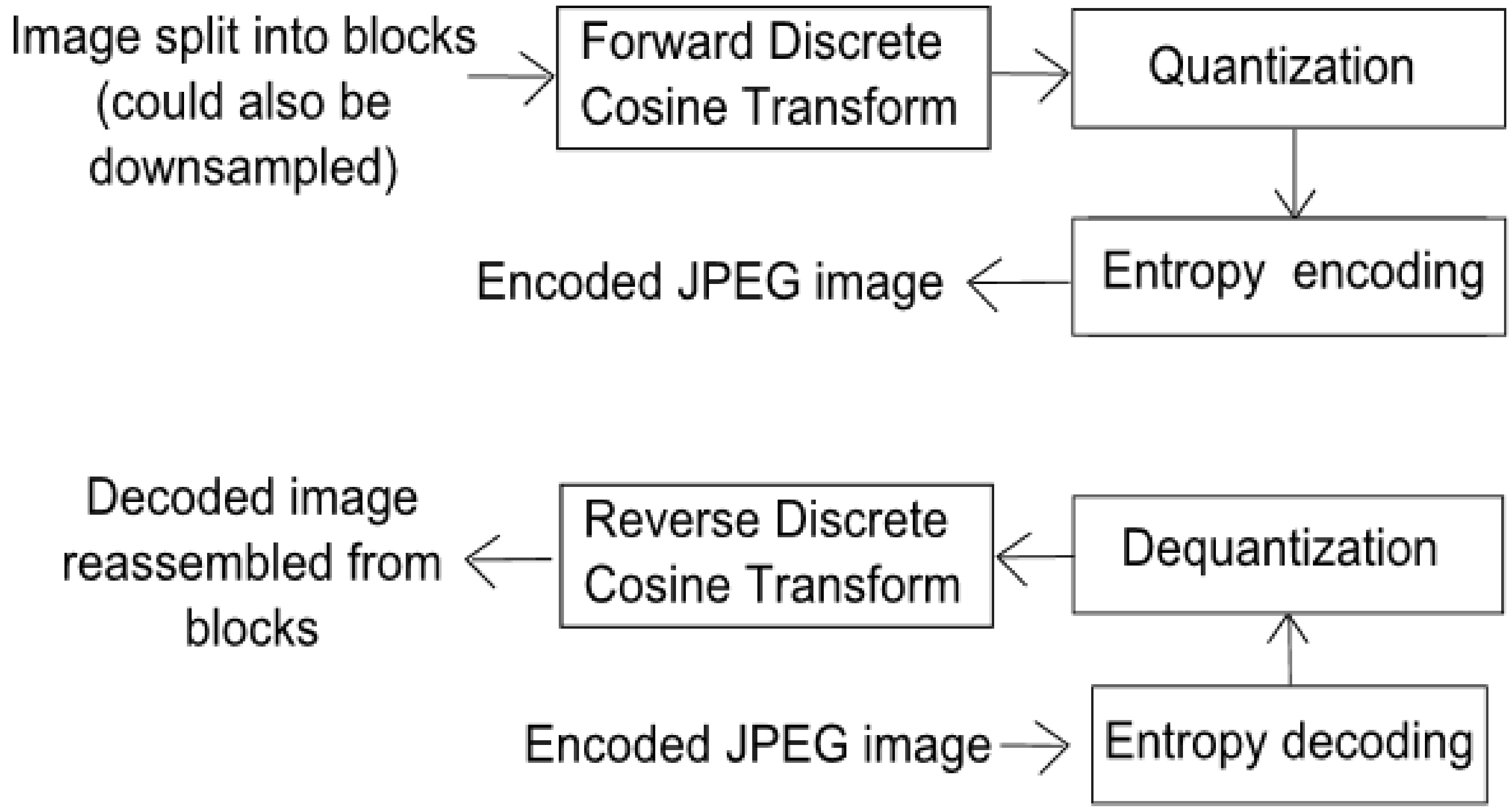


Image compression implementation

Discrete cosine transform:

$$B(k_1,k_2)=\sum_{i=0}^{N_1-1}\sum_{j=0}^{N_2-1} 4A(i,j)\cos\left(\pi\frac{k_1}{2}N_1(2i+1)\right)\cos\left(\pi\frac{k_2}{2}N_2(2j+1)\right)$$

For computational purposes it can be converted into:

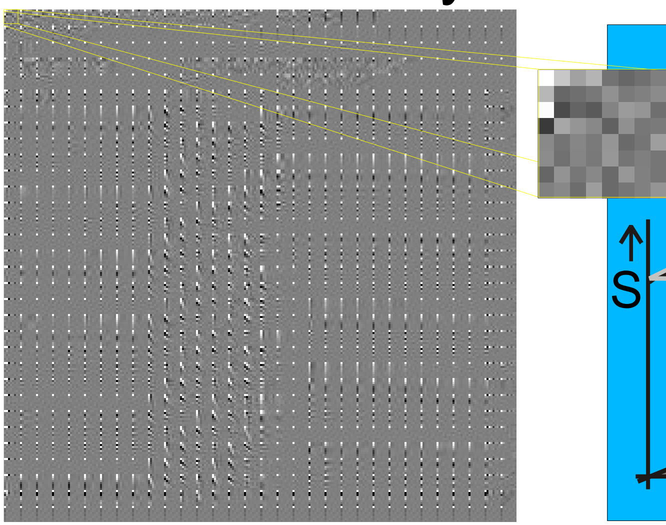
$$y_{i,j}=\sum_{ii=0}^{N_1-1}\sum_{jj=0}^{N_2-1} x_{ii,jj}k_{i,j,ii,jj}$$

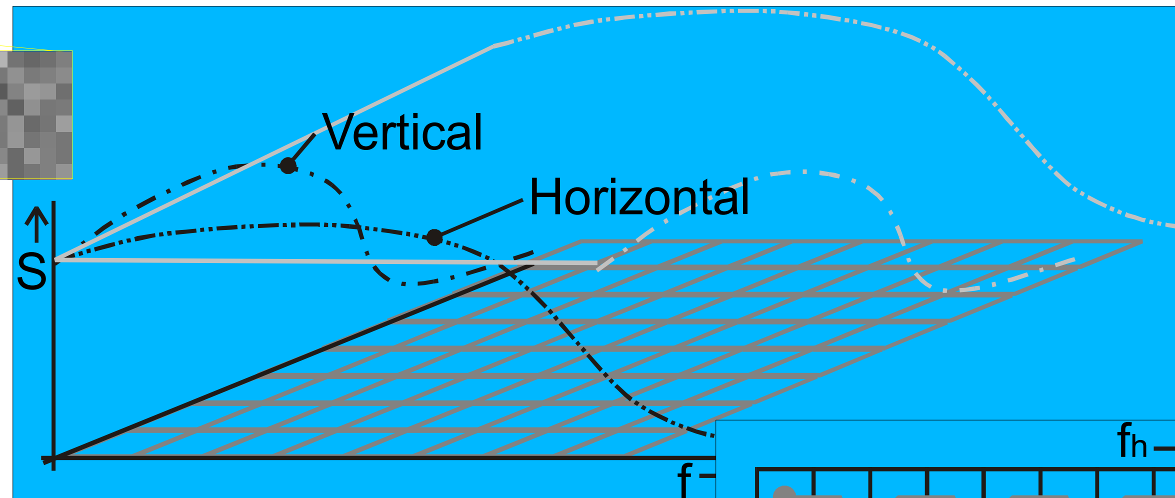
k is a tabulated set of coefficients

Inverse DCT (IDCT) is performed similarly

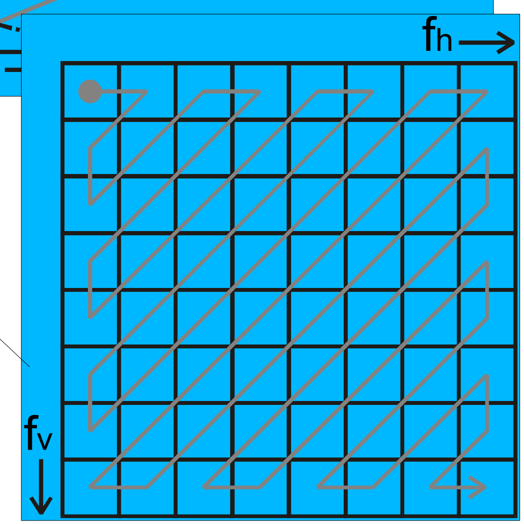
Image compression implementation

After the DCT transformation, the coefficients are divided with the sensitivity matrix (multiplied with its reciprocal value)





Then, “zigzag“ parse is performed



Bigger coefficients

Image compression implementation

- Final step in the JPEG compression is Huffmann encoding of the coefficients that result from DCT after sensitivity correction

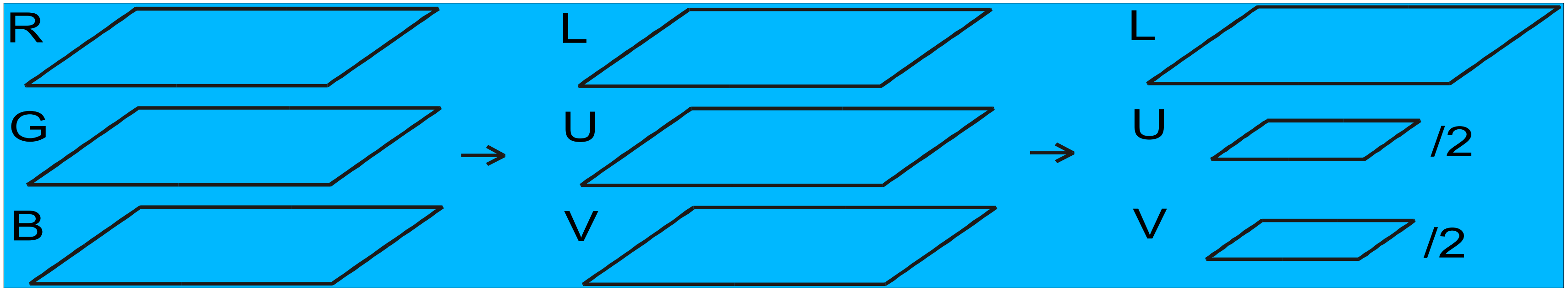
Features of the compressed images:

- Compression rates around 1:20 (depends on the data)
- Decompressed images may appear “blocky“ - under some circumstances, 8x8 blocks boundaries are visible – this drawback is, however, addressed in JPEG 2000 standard that includes wavelet based compression algorithms that reduce this type of effect
- Some features of the images may be removed during compression and decompression – the decompressed images may be unsuitable for further processing

Image compression implementation

Colour version of the JPEG compression

- First, the image is converted in a suitable colour model (preferred model is LUV type, but RGB also possible)
- In some cases (eg. in LUV), the colour components of the image may be undersampled – the reason is that the human eye is far more sensitive to the luminance changes (high frequencies)



Video compression implementation

Typical video compression is eg. MPEG (Motion Pictures Encoding Group) compression algorithms the algorithms are based on the following key ideas

- Some frames are encoded using other frame(s) (their predecessors or successors on the time line) – this is how the compression is achieved
- Some images (frames) of the video sequence are encoded fully (just like in JPEG) – the purpose is to allow replaying the video sequence from “arbitrarily“ selected location
- Some frames are encoded fully but with lower resolution – the motivation is to enable implementation of preview of the video sequence during „fast forward“ or “review“

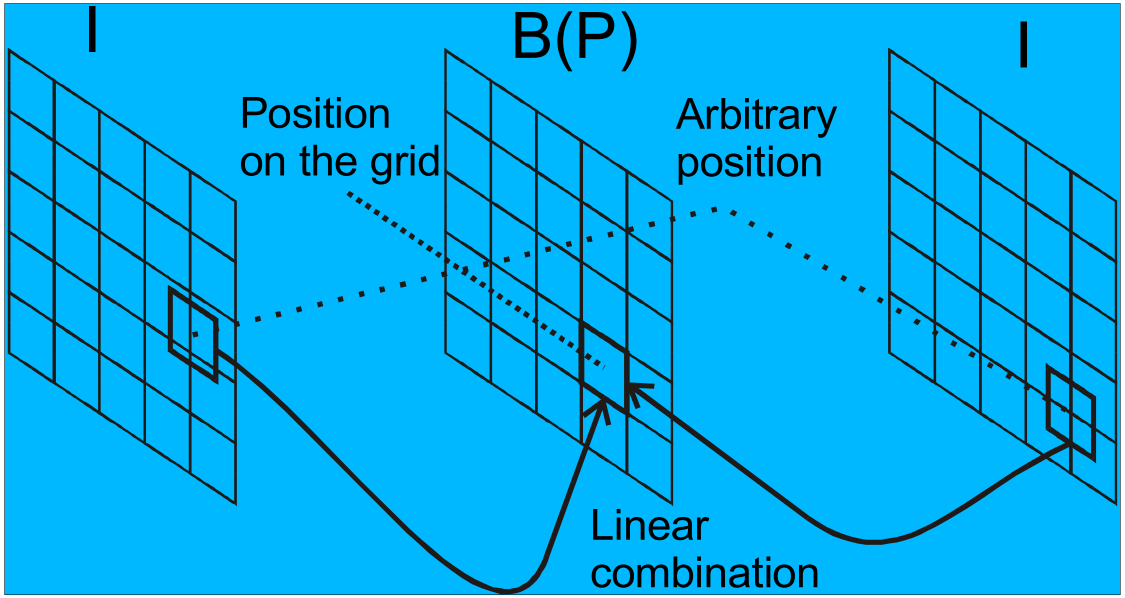
Video compression implementation

The frame types are the following

- I - Intraframe compressed frames – the fully encoded images; most encoders include these frames once in N (~50) frames
- P - Predicted frames – the frames that are based only on previous I frame
- B - Bidirectionally predicted frames – they are based on previous and future I frames
- D - DC only – means that only „DC“ coefficients of 8x8 image blocks are present – effectively 8x undersampling

Video compression implementation

The Bidirectional (B) and Predictive (P) frames are built from 16x16 blocks of pixels that are created as linear combination of 16x16 blocks from the previous and future I frames (just previous in the case of P frames); the differences (errors) are encoded similarly to JPEG



Video compression implementation

The search for the suitable blocks to use in the linear combination can be very computationally expensive, or sophisticated „motion estimation" algorithms can be used

In the case of full search in 320x240 frames, the B frame needs 320x240x320x240 tests x16x16 = 150×10^{10} MACs

Such search is not possible to really use and the optimized methods are subject of research so it would be difficult to standardize; for this reason only the process of **DECODING is standardized** in MPEG and the encoding phase is left unspecified so the developers can develop their own versions

Video compression implementation

MPEG formats:

- MPEG1 – video compression as described so far
- MPEG2 – modification that allows digital broadcast – subdivided into streams based on quality, lower quality receivers process one stream, higher more...
- MPEG4,7 – extensions that allow for better prediction through 3D object modelling and animation, the motivation is specifically video telephony through low bandwidth connections, such as standard telephone lines; the 3D objects are in this case human heads; in addition, MPEG7 defines also lips animation and annotations of video

Conclusions

- Image compression is based on knowledge of the image data statistics – redundancy
- Loosy compression (on top of the image statistics) exploits the human sight features
- Video compression uses also temporal coherence between the video frames
- Colour in image and video compression is included usually after conversion of the colour format into LUV or other similar format that encodes intensity information in one channel and colour information into other (two) channels